

Derivative Rules: Guide

The big idea. Every derivative is built from a short toolbox of rules. Your job is to **name the outermost operation** first (add, multiply, divide, or a function inside a function) and apply that rule. Then handle each piece the same way.

PICKING THE RULE

- 1 Find the outermost operation.** What would you do *last* if you evaluated the function?
- 2 Sum or constant multiple:** differentiate term by term. Constants stay out front.
- 3 Product** uv : $u'v + uv'$. **Quotient** $\frac{u}{v}$: $\frac{u'v - uv'}{v^2}$.
- 4 Function inside a function:** chain rule. Outer derivative at the inner, times the inner derivative.
- 5 Repeat inside each piece.** A product can contain a chain, and so on.
- 6 Plug in a number last.** Differentiate completely, then evaluate. Simplify only if asked.

Most common mistakes: (1) Using the power rule on 5^x (the variable is in the *exponent*: $5^x \ln 5$). (2) Product rule written as $u'v'$. (3) Quotient rule numerator in the wrong order: it is $u'v - uv'$. (4) Forgetting the inner derivative in the chain rule, as in $\frac{d}{dx} \sin(3x) = \cos(3x) \cdot 3$. (5) Differentiating $\cos$ and getting $+\sin$.

WHICH RULE?

IT LOOKS LIKE	RULE	EXAMPLE
x^n (x in the base)	Power	$x^5 \rightarrow 5x^4$
a^x (x in the exponent)	Exponential	$5^x \rightarrow 5^x \ln 5$
Two functions multiplied	Product	$x^2 e^x \rightarrow 2xe^x + x^2 e^x$
A fraction with x on the bottom	Quotient	$\frac{x}{x+1} \rightarrow \frac{1}{(x+1)^2}$
A function of a function	Chain	$\sin(x^2) \rightarrow 2x \cos(x^2)$

RULE BANK

BASIC		EXPONENTIALS & LOGS	TRIG		
Constant	$\frac{d}{dx} c = 0$	e^x	$\frac{d}{dx} e^x = e^x$	$\sin x$	$\cos x$
Power	$\frac{d}{dx} x^n = nx^{n-1}$	a^x	$\frac{d}{dx} a^x = a^x \ln a$	$\cos x$	$-\sin x$
Sum	$(f \pm g)' = f' \pm g'$	$\ln x$	$\frac{d}{dx} \ln x = \frac{1}{x}$	$\tan x$	$\sec^2 x$
Rewrite $\sqrt{x}$ and $\frac{1}{x^n}$ as powers first.		$\log_a x$	$\frac{d}{dx} \log_a x = \frac{1}{x \ln a}$	$\sec x$	$\sec x \tan x$
				$\csc x$	$-\csc x \cot x$
				$\cot x$	$-\csc^2 x$
				Every "co" function gets a minus sign.	
PRODUCT & QUOTIENT		CHAIN RULE		WORKING BACKWARD	
Product	$(uv)' = u'v + uv'$	Rule	$[f(g(x))]' = f'(g(x)) g'(x)$	Reverse power $\int x^n dx = \frac{x^{n+1}}{n+1} + C$	
Quotient	$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$	g^n	$ng^{n-1} g'$	Given f' and one point on f : undo each term, then use the point to find C .	
Quotient: "low d-high minus high d-low, over low squared."		e^g	$e^g g'$		
		$\ln g$	$\frac{g'}{g}$		
		$\sin g$	$\cos g \cdot g'$		

Eight problems that cover the exam

Name the outermost operation, then apply its rule.

Walkthrough 1: Exponential and log

TERM BY TERM

Differentiate $f(x) = 3e^x - 2 \cdot 5^x + 4 \ln x$.**1 E^x TERM** e^x is its own derivative: $3e^x$.**2 5^x TERM** Variable in the exponent: $5^x \ln 5$, so $-2 \cdot 5^x \ln 5$.**3 LN TERM** $\frac{4}{x}$ **4 COMBINE** $f'(x) = 3e^x - 2 \cdot 5^x \ln 5 + \frac{4}{x}$

Walkthrough 2: Product with trig

U'V + UV'

Differentiate $f(x) = x^2 \sin x$.**1 NAME** $u = x^2, v = \sin x$. So $u' = 2x, v' = \cos x$.**2 APPLY** $u'v + uv' = 2x \sin x + x^2 \cos x$ **3 ANSWER** $2x \sin x + x^2 \cos x$

Walkthrough 3: Quotient

LOW D-HIGH...

Differentiate $f(x) = \frac{\ln x}{x^2}$.**1 NAME** $u = \ln x, v = x^2$. So $u' = \frac{1}{x}, v' = 2x$.**2 APPLY** $\frac{\frac{1}{x} \cdot x^2 - \ln x \cdot 2x}{x^4} = \frac{x - 2x \ln x}{x^4}$ **3 SIMPLIFY** Cancel an x : $\frac{1 - 2 \ln x}{x^3}$

Walkthrough 7: Name the rule

QUICK DRILL

Which rule starts each derivative: x^5 vs 5^x , $x^2 e^x$, e^{x^2} , $\frac{x+1}{x-3}$?**1 X⁵ VS 5^x** x^5 : power, $5x^4$. 5^x : exponential, $5^x \ln 5$.**2 PRODUCT?** Two functions multiplied: **product**.**3 CHAIN?** Inside function x^2 : **chain**, $2xe^{x^2}$.**4 QUOTIENT?** Fraction with x on the bottom: **quotient**.

Walkthrough 9: Product and chain together

RULES COMBINE

Differentiate $f(x) = e^{2x} \cos x$.**1 OUTER** Two factors multiplied: **product**. $u = e^{2x}, v = \cos x$.**2 INNER** $u' = 2e^{2x}$ (chain: inside is $2x$). $v' = -\sin x$.**3 APPLY** $u'v + uv' = 2e^{2x} \cos x - e^{2x} \sin x$ **4 FACTOR** $e^{2x}(2 \cos x - \sin x)$

Walkthrough 4: Chain rule

OUTSIDE, THEN INSIDE

Differentiate (a) $\sin(3x^2)$, (b) $(2x+1)^5$, (c) $\ln(x^2+1)$.**1 (A)** Outer sin, inner $3x^2$: $\cos(3x^2) \cdot 6x = 6x \cos(3x^2)$ **2 (B)** Outer power, inner $2x+1$: $5(2x+1)^4 \cdot 2 = 10(2x+1)^4$ **3 (C)** Outer ln, inner x^2+1 : $\frac{2x}{x^2+1}$

Walkthrough 5: Chain rule from a table

PLUG IN CAREFULLY

 $h(x) = f(g(x))$. Given $g(1) = 3, g'(1) = 2, f(3) = 5, f'(3) = -4$. Find $h'(1)$ and $p'(1)$ for $p(x) = [g(x)]^2$.**1 CHAIN** $h'(1) = f'(g(1)) \cdot g'(1)$ **2 PLUG IN** $g(1) = 3$, so $f'(3) \cdot 2 = -4 \cdot 2 = -8$ **3 POWER** $p'(1) = 2g(1)g'(1) = 2(3)(2) = 12$ **4 WATCH** f' is evaluated at $g(1)$, **not** at 1.

Walkthrough 6: Tangent line

PRODUCT, THEN LINE

Find the tangent line to $f(x) = xe^x$ at $x = 1$.**1 DERIVATIVE** $f'(x) = 1 \cdot e^x + x \cdot e^x = e^x(1+x)$ **2 SLOPE** $f'(1) = e \cdot 2 = 2e$ **3 POINT** $f(1) = e$, so the point is $(1, e)$.**4 WRITE IT** $y - e = 2e(x - 1)$, so $y = 2ex - e$

Walkthrough 8: Work backward

REVERSE THE POWER RULE

Given $f'(x) = 6x^2 - 4x + 1$ and $f(1) = 5$, find $f(x)$.**1 REVERSE** Raise each exponent by 1, divide by the new one: $2x^3 - 2x^2 + x + C$.**2 USE F(1)** $2 - 2 + 1 + C = 5$, so $C = 4$.**3 ANSWER** $f(x) = 2x^3 - 2x^2 + x + 4$ **4 CHECK** Differentiate: $6x^2 - 4x + 1$ ✓